ECE 1212 – Electronic Circuit Design Laboratory
Experiment #1 – Frequency Response of Operational Amplifiers

Introduction

You are to design several linear amplifiers with specified gains and bandwidths. You will use the 741 operational amplifier (op-amp) as the principal component in your designs. Use $\pm$ 12V from the power supply for your op-amp power, or lower input voltages to achieve desired gains if power supply is constrained. The lab will be split into 3 parts: I. Amplifier Characterization, II. Filter Frequency Response, III. Filter Modification

Deliverables

1. Prelab 1A (part I – 2 prompts)
2. Prelab 1B (part II – 3 prompts, part III – 2 prompts)
3. Lab Report

Part I – Amplifier Characterization

1. Design, build and test non-inverting amplifiers with voltage gains of $A_V = 1$ and $A_V = 10$.

   In the Pre-Lab (1A):
   Provide schematic diagrams for each design, using as few components as necessary. Predict what the 3 dB bandwidth for each amplifier will be, making use of equations from the lecture notes. Comment on the applications for the $A_V = 1$ amplifier.

   In the Laboratory:
   (a) Construct each of the non-inverting amplifiers, using the oscilloscope as the load. Measure the low-frequency gain and the 3 dB bandwidth, and compare to the values expected from your design calculations.
   (b) Use your measurements of the 3-dB bandwidths of the amplifiers to calculate f_t, the unity-gain bandwidth of the op-amp. Compare the two values you obtain for f_t to each other and to the nominal value provided by the data sheet for the 741 op-amp.
   (c) Measure and report on the magnitude of the frequency response for each amplifier.

2. Input impedance of the non-inverting amplifier at $f = 1$ kHz.

   In the Pre-Lab (1A):
   Show a schematic diagram of the circuit used for measuring the input impedance. Referring to the lecture notes, explain the testing procedure and how you will calculate the input impedance from your measurements. Predict what you expect the input impedances to be for the two non-inverting amplifiers.

   In the Laboratory:

Try to measure the input impedances of the non-inverting amplifiers at a frequency of 1 kHz. Compare the values you obtain to what is expected from the lecture notes. If there is a significant discrepancy between the measured and expected values, you will need to explain this in the Discussion section of your report. Why is it difficult to measure the input impedance using this testing procedure?

Part II – Filter Frequency Response

1. Design, construct and test an inverting amplifier which:
   - has flat (within 3dB) frequency response up to 20kHz, and
   - has an input impedance of 1 kΩ (for low frequencies, i.e. *f* close to 0).

   Your amplifier will drive a 3.3 kΩ load resistor. Design for the highest possible gain compatible with these criteria.

   In the Pre-Lab (1B):
   Provide a schematic diagram of your design, and show all of the calculations to determine the component values. The design calculations will require you to have a value for f_t, the unity-gain bandwidth of the op-amp. Use one of the measured values for f_t that you obtained in Part I.

   In the Laboratory:
   (a) Construct and test your circuit, determine the low-frequency gain and the 3dB bandwidth, and compare with the design specifications.
   (b) Use the oscilloscope or Sweep Demo to measure the frequency response, prepare a Bode magnitude plot and compare with the Bode plot predicted by the closed-loop gain equation from the lecture notes. You should include a wide range of input frequencies in your tests, including a dense sampling of frequencies below 3dB bandwidth, and a coarse sampling of frequencies up to and including the unity-gain bandwidth of the op-amp.

2. Modify your design so that it can compensate for DC components of the input signal, and produce an output signal with zero DC value. Compensation should be accomplished with a potentiometer or a decade resistor. You may assume that the maximum input voltage swing is ± 200 mV peak-to-peak, but the average DC level of the signal may be anywhere between -1.0 V to +1.0 V.

   In the Pre-Lab (1B):
   Provide a schematic diagram of your modified design, and explain how the DC compensation works. Why is it important to provide this compensation?

   In the Laboratory:
   Construct and test your circuit, and demonstrate the ability to compensate for the DC level of the input signal. Record what happens to the output as you raise the DC input voltage amplitude or vary the DC compensation, and document this in your lab report.

The DC level available from the signal generator depends on the amplitude of the signal. It may not be possible to actually produce a DC signal level as large as 1.0 V with an input voltage swing as small as 200 mV peak-to-peak. If you are not able to produce a signal with a DC offset of ± 1 V, use a signal with as large of a DC offset as is available from the generator.

3. Measure the input impedance of your inverting amplifier using the similar to Part I.

   In the Pre-Lab (1B):
   Show a schematic diagram of the circuit used for measuring the input impedance, and explain the testing procedure.

   In the Laboratory:
   Use this approach to measure the input impedance with the signal generator set to 100 Hz, 1 kHz, 5 kHz, 15 kHz, and several higher frequencies. Make a plot of the variation of the input impedance with frequency, and compare to what is expected from the equations that appear in the lecture notes. Explain why this happens in your report.